

Bibliography

- [1] Bourbaki, *Algèbre*. Springer, 2007.
- [2] M. F. Atiyah and I. G. Macdonald, *Introduction to commutative algebra*. CRC Press, 2018.
- [3] R. Bott and L. W. Tu, *Differential forms in algebraic topology*, vol. 82. Springer Science & Business Media, 2013.
- [4] D. A. Buchsbaum, “Exact categories and duality,” *Transactions of the American Mathematical Society* **80** no. 1, (1955) 1–34.
- [5] A. Grothendieck, “Sur quelques points d’algèbre homologique,” *Tohoku Mathematical Journal, Second Series* **9** no. 2, (1957) 119–183. English translation available at <https://www.math.mcgill.ca/barr/papers/gk.pdf>.
- [6] F. Peter, “Abelian Categories: An Introduction to the Theory of Functors,”.
- [7] J.-L. Verdier, “Catégories dérivées Quelques résultats (état 0),” in *Cohomologie Etale: Séminaire de Géométrie Algébrique du Bois-Marie SGA 4 1/2*, pp. 262–311. Springer, 1977.
- [8] C. A. Weibel, *An introduction to homological algebra*. No. 38. Cambridge university press, 1994.
- [9] D. Happel, *Triangulated categories in the representation theory of finite dimensional algebras*, vol. 119. Cambridge University Press, 1988.
- [10] S. MacLane, “Henri Cartan and Samuel Eilenberg, Homological Algebra,”.
- [11] T. Stacks project authors, “The Stacks project.” <https://stacks.math.columbia.edu>, 2025.
- [12] C. A. Weibel, *The K-book: An Introduction to Algebraic K-theory*, vol. 145. American Mathematical Soc., 2013.
- [13] M. Atiyah, *K-theory*. CRC press, 2018.
- [14] H. Miller, “Leray in Oflag XVIII: The origins of sheaf theory, sheaf cohomology, and spectral sequences,”.
- [15] J.-P. Serre, “Faisceaux algébriques cohérents,” *Annals of Mathematics* **61** no. 2, (1955) 197–278.
- [16] A. Grothendieck, “Éléments de géométrie algébrique: I. Le langage des schémas,” *Publications Mathématiques de l’IHÉS* **4** (1960) 5–228.
- [17] A. Verma, “Local aspects of topological quantization and the Wu-Yang Monopoles,” *arXiv:2406.18799* (2024) .
- [18] T. Y. Chow, “You could have invented spectral sequences,” *Notices of the AMS* **53** (2006) 15–19.
- [19] A. HATCHER, “Algebraic Topology,” <http://www.math.cornell.edu/~hatcher/AT/ATpage.html> (2002) .
- [20] J. McCleary, *A user’s guide to spectral sequences*. No. 58. Cambridge University Press, 2001.
- [21] M. Dubois-Violette, “The Weil-BRS algebra of a Lie algebra and the anomalous terms in gauge theory,” *Journal of Geometry and Physics* **3** no. 4, (1986) 525–565.
- [22] R. Vakil, “The rising sea: Foundations of algebraic geometry,”. available at <https://math.stanford.edu/~vakil/216blog/FOAGjul2724public.pdf>.
- [23] W. Browder, “Torsion in H-spaces,” *Annals of Mathematics* **74** no. 1, (1961) 24–51.
- [24] A. Weil, *Foundations of algebraic geometry*, vol. 29. American Mathematical Soc., 1946.

BIBLIOGRAPHY

- [25] R. Hartshorne, *Algebraic geometry*, vol. 52. Springer Science & Business Media, 2013.
- [26] D. Mumford, *The red book of varieties and schemes: includes the Michigan lectures (1974) on curves and their Jacobians*, vol. 1358. Springer, 2004.
- [27] D. Dummit and R. Foote, *Abstract Algebra*. Wiley, 2003.
- [28] D. Hilbert, “Über die vollen Invariantensysteme,” in *Algebra• Invariantentheorie Geometrie*, pp. 287–344. Springer, 1970.
- [29] H. Matsumura, *Commutative ring theory*. No. 8. Cambridge university press, 1989.
- [30] C. Birkar, “Lectures on birational geometry,” *arXiv preprint arXiv:1210.2670* (2012) .